

AFWERX Prime Innovative Capabilities Opening (ICO)
Electrification of Fixed Wing Aircraft - Call 0002

Issued as a part of ICO Number	FA228024R0007		
Open Date:	20 March 2026	Close Date:	20 April 2026 11:59 EDT
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Performance Restriction:	Restricted to US based firms only (Export Controlled)	Status: Open	Accepting Solutions?: Yes

Table 1: Call 0002 at a glance

1. Introduction

Under the authority of 10 U.S.C. § 4022, the AFWERX Prime (AFRL/RGP) Office is soliciting prototype projects from nontraditional and traditional defense contractors. These projects are intended to help the Government assess and accelerate emerging electrification and hybrid-electric propulsion technologies for fixed-wing aircraft in partnership with industry.

This Call seeks to establish a collaborative strategy with industry and investors to accelerate the maturation and demonstration of promising technologies that may provide operational utility to the Government, as well as potential commercial market success. As technologies mature toward certified commercial or military operations, the Government may identify opportunities for early adoption, with the potential for procurement and fielding through this ICO.

2. Background

Electrification and hybrid-electric propulsion technologies offer the potential to improve fixed-wing aircraft performance, reduce operating and sustainment costs, lower acoustic signatures, and decrease logistics burdens—particularly in distributed and austere operating environments.

While prior AFWERX Prime efforts have focused heavily on vertical lift and eVTOL platforms, a substantial portion of current and future Department of the Air Force (DAF) mission sets rely on fixed-wing aircraft. Industry has begun advancing electric and hybrid-electric propulsion

solutions for fixed-wing platforms, including supplemental type certification (STC) efforts for commercially derived aircraft. These developments create an opportunity for the Government to better understand performance tradeoffs, operational relevance, and transition pathways for electrified fixed-wing aircraft concepts.

This Call emphasizes solutions that are sufficiently mature to support credible demonstration and the generation of quantitative performance data within the period of performance. Concepts that rely primarily on early-stage research without a clear and executable path to demonstration may be less competitive.

3. Problem Statement

AFWERX Prime focuses on accelerating emerging aerospace capabilities that improve operational effectiveness, reduce sustainment burden, and enable Agile Combat Employment (ACE). While significant progress has been made in electric and hybrid-electric vertical takeoff and landing (eVTOL) aircraft, the Department of the Air Force and other DoD organizations continue to rely on a wide range of non-eVTOL fixed-wing aircraft for missions where electrification or hybridization may provide meaningful advantages.

Industry has initiated efforts to develop and certify electric and hybrid-electric powertrains for fixed-wing aircraft, including STC efforts for commercially derived platforms such as the Cessna Caravan family. These efforts indicate growing technical maturity; however, the cost, performance, integration, and operational tradeoffs of electrified fixed-wing aircraft relative to conventional military and commercial baselines are not yet well characterized for military-relevant use cases.

Many fixed-wing aircraft used across DAF and DoD missions continue to rely on conventional propulsion architectures that impose significant fuel, logistics, sustainment, and acoustic constraints—particularly in distributed, austere, or contested environments. Electrified or hybridized fixed-wing aircraft may offer improvements in efficiency, operational resilience, and mission flexibility, but require demonstration-based evidence to support future operational, acquisition, and certification decisions.

Accordingly, the Government seeks to evaluate and mature electrification approaches for fixed-wing aircraft through modeling and simulation, ground testing, prototype demonstration, and/or full-scale aircraft demonstration, as appropriate to the maturity of the proposed solution. Proposed efforts may address propulsion architectures, power and energy management, thermal management, system integration, and aircraft-level performance impacts. Offerors are encouraged to clearly articulate baseline assumptions and comparison methodologies to enable meaningful assessment of performance improvement and residual risk.

4. Objectives

The objectives of this Call are to:

- Demonstrate improved performance of electrified or hybridized fixed-wing aircraft relative to conventional fixed-wing baselines, supported by quantitative data
- Advance technology maturity and reduce system and integration risk
- Assess applicability of electrification technologies to emerging military mission needs, including the integration of military-relevant sensors or payloads where appropriate
- Generate credible, decision-quality technical data to inform future transition, acquisition, or rapid fielding considerations

5. Prototype Lifecycle Procedure

This Call will be executed using a phased prototype lifecycle approach under the authority of 10 U.S.C. § 4022. Submission requirements, evaluation criteria, and solicitation mechanics are governed by the Call Procedures section and are incorporated by reference. The Government reserves the right to select one, multiple, or no offerors at any stage.

a. Solicitation Phase

During the Call open period, offerors shall submit a Pitch Deck in accordance with the Call Procedures. Pitch Decks should describe the proposed electrification or hybrid-electric fixed-wing approach, current technology maturity, demonstration strategy, anticipated schedule, and plan to generate quantitative performance data relative to a defined baseline.

b. Evaluation and Selection

Pitch Decks will be evaluated based on eligibility and the published basis of evaluation, including technical feasibility, importance to the Government, technology maturity, and the credibility of the proposed data-generation and demonstration approach.

c. Contract Award

Based on evaluation results, the Government may invite selected offerors to submit a Request for Proposal (RFP). Following proposal evaluation and negotiations, the Government may award one or more Other Transactions for Prototype.

d. Test and Demonstration Planning

Awardees will conduct technical and demonstration planning appropriate to the maturity of the proposed solution. Planning activities may include baseline definition, test objectives, demonstration configuration, data collection methods, safety considerations, and risk mitigation strategies.

e. Demonstration Execution (as applicable)

Awardees may execute demonstrations consistent with their approved plans. Demonstrations may include modeling and simulation, component or subsystem testing, integrated ground testing, and/or prototype or aircraft-level demonstration, including flight testing where appropriate. Demonstrations may be conducted at contractor facilities or other approved locations.

f. Data Collection and Reporting

Awardees shall collect and deliver technical data, analyses, and reports sufficient to substantiate claimed performance improvements, technology maturity, and residual risk. These results may be used by the Government to inform future investment, transition, acquisition, or certification considerations.

6. Definitions

- a. Definitions in the ICO FA228024R0007 are incorporated by reference.
- b. “Open Period” refers to the date that a Call is open for offerors to submit solutions. Solutions will be evaluated as soon as practicable (i.e., as they are received) and awards may be issued during the open period. Evaluations and awards may be made after the closing date for those submissions received prior to the closing date and time provided within the Call. Solutions submitted after the closing date will not be considered

7. Anticipated Deliverables

Deliverables should be commensurate with the technical maturity of the proposed solution. Offerors are encouraged to propose deliverables that demonstrate increasing levels of system integration, validation, and quantitative performance data as maturity increases. Solutions at higher levels of maturity are expected to emphasize demonstration-based evidence.

At a minimum, anticipated deliverables may include the following, as appropriate to the proposed effort.

Deliverables Table

Deliverable	Description	Acceptance Criteria
Program Kickoff & Reviews	Kickoff and periodic technical/management reviews aligned to milestones	Reviews conducted; materials delivered
Baseline Definition & Metrics	Definition of baseline aircraft/configuration and comparison methodology	Baseline and metrics documented
System Architecture & Integration	Description of propulsion, energy, thermal, and integration approach	Documentation supports feasibility assessment
Engineering Analysis & M&S	Analyses supporting aircraft-level performance claims	Results traceable to baseline
Test & Demonstration Plan (if applicable)	Ground and/or flight demonstration objectives and data plan	Plan delivered and aligned to milestones
Ground Test Data & Analysis (if applicable)	Subsystem or integrated ground test data	Data supports performance claims
Prototype / Aircraft Demonstration Results (if applicable)	Results from prototype or aircraft-level demonstration	Results support milestone assessment
Final Technical Report	Summary of results, data, limitations, and residual risks	Report delivered and accepted

Not all deliverables are required for every award. Deliverables should be appropriate to the maturity and scope of the proposed effort.

8. Period of Performance

Awards issued under this Call are expected to have a maximum period of performance of 24 months.

9. Place of Performance

Work shall be performed primarily at contractor facilities, unless otherwise proposed and approved.

10. Evaluation Emphasis

In evaluating submissions, the Government may place greater emphasis on solutions that demonstrate higher levels of technical maturity, clearly defined performance baselines, and a

credible approach to producing quantitative performance data through modeling, ground testing, or flight demonstration within the period of performance.

11. Call Procedures

- a. Pitch Deck submission requirements, basis of evaluation, Request for Proposal procedures, non-Government advisor disclosures, and standard AFWERX terms and conditions are incorporated by reference and remain unchanged.
- b. Open Period - this Call is an open solicitation with a closing date of 11:59 PM EDT 04/18/2026.
- c. Step 1 - Pitch Deck Submission - Submit a not-to-exceed (NTE) 10 slide Pitch Deck via email to afrl.primeicocall002.ext@afrl.af.mil. Pitch Decks should generally contain the following information:
 - i. In the event that an offeror submits a Pitch Deck exceeding 10 slides, only Slides 1 through 10 will be considered by AFWERX evaluators. Excess slides will be removed prior to review and will not be considered by AFWERX evaluators.
 - ii. Any other correspondence must be submitted electronically to the Agreements Officer point(s) of contact identified in Table 1 located at the beginning of this Call.
 - iii. Submissions sent through other communication channels, or after the open period has closed will not be considered.
 - iv. The Government will not pay submitters for the costs associated with Pitch Deck submissions, site visits, or developing materials for proposals, unless otherwise stipulated within this announcement.
 - v. Unnecessarily elaborate Pitch Decks are not desired. Use of diagram(s) or figure(s) to depict the essence of the proposed solution is strongly encouraged.
 - vi. Foreign Participation – Foreign persons or foreign owned firms are precluded from participation.
 - vii. Pitch Decks may only provide one solution meeting the objectives described in this Call. However, offerors may submit multiple Pitch Decks in response to a single Call.
 - viii. All Pitch Decks shall be unclassified. Pitch Decks, Site Visit, and Proposals containing data that is not to be disclosed to the public for any purpose or used by the Government except for evaluation purposes shall include the following statement –

“This [select one: Solution Submission or Proposal] includes proprietary information that shall not be disclosed outside the Government, except to

non-Government personnel for evaluation purposes, and shall not be duplicated, used, or disclosed -- in whole or in part -- for any purpose other than to evaluate this submission. If, however, an agreement is awarded to this Offeror as a result of -- or in connection with -- the submission of this data, the Government shall have the right to duplicate, use, or disclose the data to the extent agreed upon by both parties in the resulting agreement. This restriction does not limit the Government's right to use information contained in this proprietary information if it is legally obtained from another source without restriction. The data subject to this restriction are contained in sheets [insert numbers or other identification of sheets]”

d. Step 1 - Pitch Deck: Basis of Evaluation

- i. Eligibility: The entity must be registered via SAM.gov and be eligible to receive federal awards. Note: Awards under this effort are not anticipated to be assistance instruments i.e. grants. The entity must be eligible for prototype award under 10 U.S.C. § 4022(d).
- ii. Technical Feasibility: Pitch Deck briefs must adequately demonstrate the ability to achieve stated technical objectives and address the offerors understanding of the needs/requirements listed within the specific call. The Pitch Deck must, at a minimum, describe the solution's ability to meet the stated objectives in this Call or provide enough information to describe a path by which the solution may be adapted to meet the objective/requirement.
- iii. Importance to the Government: The Pitch Deck must adequately demonstrate its perceived importance to the Government. This should be related to stated missions, objectives, characteristics, relevant operational imperative or perceived fit within a technology area based on the vendor's understanding of the technology being proposed.

- e. The Government may elect to use external market research in the evaluation of an offeror's viability. The Government holds the right to select all, some, or none of the solution submissions submitted in response to the applicable call.

f. Step 2 - Request For Proposal (RFP): Submission Content

- i. The second and final Step of the ICO process is the RFP. Based upon the results of Step 1, the Government may issue an RFP. If that occurs, an offeror, or offerors, will be invited to develop and submit a written proposal as well as negotiate appropriate terms and conditions, as applicable, governing the proposed solution and/or prototype. At this stage, the selected offerors may discuss proposal development details during the proposal writing process with their assigned Government subject matter expert(s). Each proposal submitted shall consist of two sections: Section 1 shall provide the technical proposal; Section 2 shall address the price/cost/schedule

portions of the proposal. Additionally, offerors are invited to provide their own internal terms and conditions that may be consolidated with those of the Government. This includes Service License Agreements (SLA) and/or User License Agreements (ULA).

- ii. Instructions to Offerors will be provided in each RFP.
- iii. Prior to the award of any contract or agreement, Submitters may be required to submit a DD 2345; Militarily Critical Technical Data Agreement.

12. Non-Government Advisors

- a. Offerors are advised that data submitted to the Government in response to this solicitation may be released to individuals who work for the following Government support contractors:
 - i. Modern Technology Solutions, Inc. (MTSI)
 - ii. Peerless Technologies

13. Disclaimer

This Call shall not be construed as a contractual or award commitment on the part of the Government. An acceptable proposal, successful negotiations, and the availability of appropriate funding are prerequisites to the issuance of any agreement. The Government reserves the right to cancel this solicitation, in whole or in part, at any time. The Government has no obligation to reimburse offerors for any costs associated with responding to this Call.